MIL-C-13720B(ME)
7 November 1983
SUPERSEDING
MIL-C-13720A(ME)
9 January 1967

MILITARY SPECIFICATION

CLAMP, BEAM, CARGO LASHING

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cargo lashing beam clamp.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Command, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 675	- Steel Bars, Carbon, Hot-Wrought, Special Quality, Mechanical Properties.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The clamp shall be as shown in figure 1 and as specified herein. The total weight shall not exceed 8-1/2 pounds.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. The clamp and components shall be hot-rolled carbon steel conforming to ASTM A 675 (see 6.4).

3.3.1 Material deterioration and control. The clamp shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Strength. The clamp shall withstand a proof load of 5,000 pounds without permanent deformation, cracks, or splits in any component. The safe working load (SWL) shall be 2,500 pounds.

3.5 Body. The body shall be one piece without welding. The throat shall be constructed so that ships' structural members having webs 1/4 to 13/16 inch thick and flanges 2 to 4-3/8 inches nominal width can be received and clamped without interference between the clamping device and the fillet of the structural member.

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3.6 Wedge. The length of the tapered portion of the wedge shall be not more than 8 inches. The tapered portion shall have an angle of taper not more than 5 degrees and shall provide a reserve of not less than 1/16 inch of clamping action when the clamp is attached to 1/4 inch thick flanges.

3.7 Eye. The eye may be integral with, or attached to, the body and may be rigid or movable. The eye shall accommodate 1 inch diameter wire rope or 2 inch steel strapping interchangeably.

3.8 Finish. The clamp and component shall be self-colored.

3.9 Identification marking. The clamp shall be legibly die stamped with the SWL and the manufacturer's name or trademark in accordance with MIL-STD-130.

3.10 Workmanship. The clamp and component shall be free from defects such as burrs, fins, cracks, flaws, sharp edges, and other harmful extraneous material. All edges shall be rounded or chamfered. The surfaces of all parts to be welded shall be free from rust, scale, grease and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 Examination. The first article clamp shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article clamp shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The clamps shall be examined as specified herein, for the following defects:

Major

101. Clamp is not shown in figure 1 and as specified.
102. Materials not as specified.
103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatments.
106. Components missing or not as specified.
107. Tolerances not as specified.
108. Workmanship not as specified.

Minor

201. Finish not as specified.
202. Identification marking missing, incomplete, or illegible.

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4.5.2 Proof test. Fasten the clamp to a channel beam or in a testing machine. Apply a 5,000-pound load to the eye. Nonconformance to 3.4 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109.	Preservation not applied as specified for level A.
110.	Components not secured toghether as specified for level A.
111.	Boxes not as specified for level A or B.
112.	Quantities packed together exceed the size or weight limitations of the box for level A or B.
113.	Clamps not blocked within the box to prevent movement as specified for level A or B.
114.	Preservation and packing not in accordance with the reference document as specified for commercial.
115.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Unpainted surfaces of the clamp shall be coated with type P-1 preservative conforming to the applicable specification listed in and shall be applied in accordance with MIL-P-116. After the preservative has dried, the components shall be secured together with tape conforming to PPP-T-97, type II, class optional.

5.1.2 Commercial. Clamps shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

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5.2.1 Level A. Clamps, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. The clamps shall be blocked within the box to prevent movement in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Clamps, preserved as specified in 5.1, shall be packed together as specified in 5.2.1, except the box shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, V15c, style optional or boxes conforming to PPP-B-640, class 2, style optional, may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. The clamps shall be blocked within the box to prevent movement in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Clamps, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The clamp is intended for use aboard ships and is designed to grip the flange or leg of a structural beam by wedging action and thereby provide a secure anchoring point for lashing cargo.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2, 4.3, and 6.3).

c. Degree of preservation and degree of packing required (see 5.1 and 5.2)

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

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6.4 Recycled material. It is encouraged that recycled material be used when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
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Review activity:
DLA - IS

Preparing activity:
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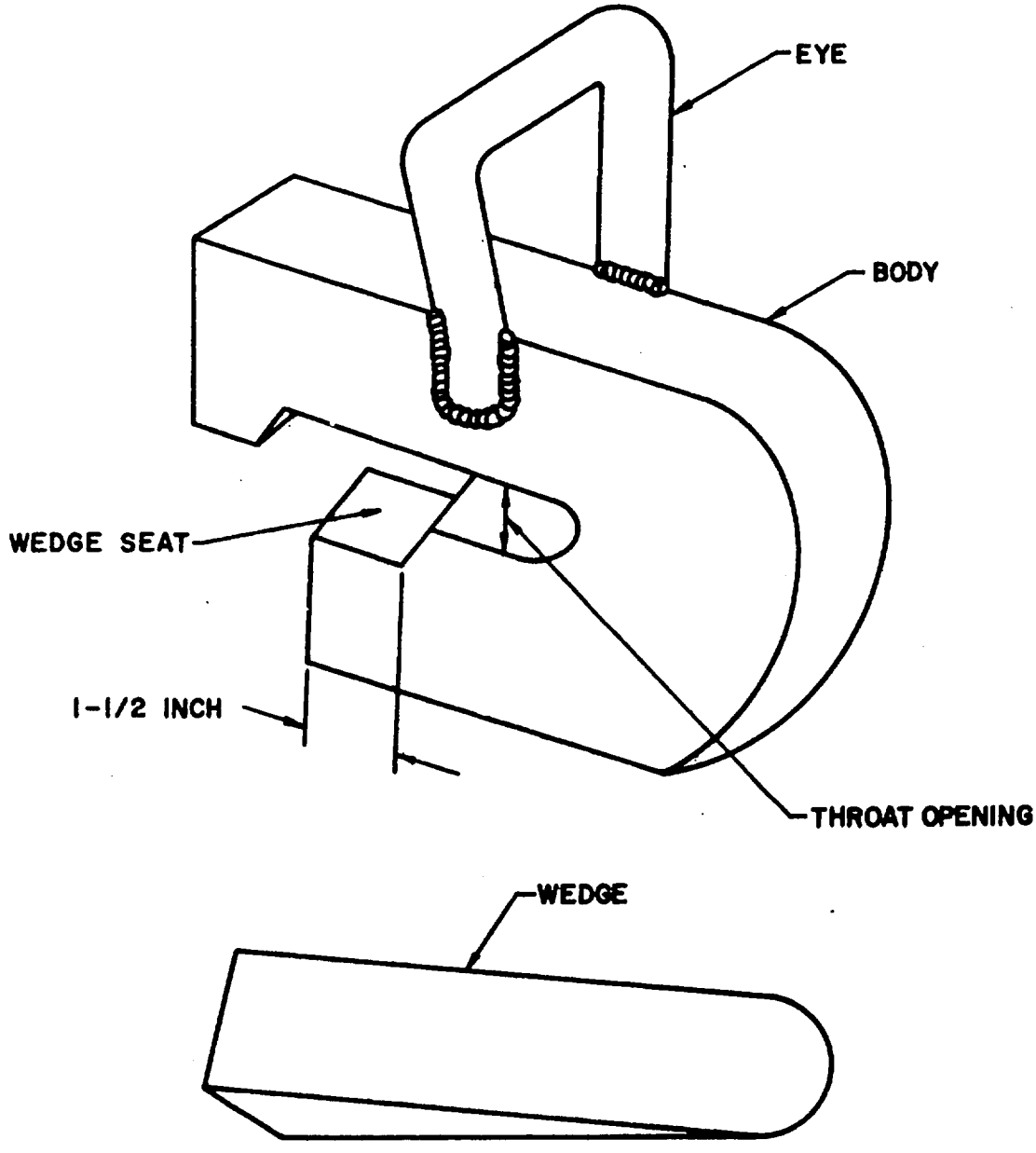


FIGURE 1. Clamp, beam, lashing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-13720B(ME)	Clamp, Beam, Cargo Lashing

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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